gravitationally. I shall call such material *exotic* because, as we shall see, it is quite different from any material that any human has ever yet met.

Second, I discovered that, just as the required exotic material must push the wormhole's walls outward, so also, whenever a beam of light passes through the material, the material will gravitationally push outward on the beam's light rays, prying them apart from each other. In other words, the exotic material will behave like a "defocusing lens"; it will gravitationally defocus the light beam. See Box 14.1.

Third, I learned from the Einstein field equation that, in order to gravitationally defocus light beams and gravitationally push the wormhole's walls apart, *the exotic material threading the wormhole must have a negative average energy density, as seen by a light beam traveling through it.* This requires a bit of explanation. Recall that gravity (spacetime curvature) is produced by mass (Box 2.6) and that mass and energy are equivalent (Box 5.2, where the equivalence is embodied in Einstein's famous equation $E = Mc^2$). This means that gravity can be thought of as produced by energy. Now, take the energy density of the material inside the wormhole (its energy per cubic centimeter), as measured by a light beam—that is, as measured by someone who travels through the wormhole at (nearly) the speed of light—and average that energy density along the light beam's trajectory. The resulting averaged energy density must be negative in order for the material to be able to defocus the light beam and hold the wormhole open-that is, in order for the wormhole's material to be "exotic."[3]

This does not necessarily mean that the exotic material has a negative energy as measured by someone at rest inside the wormhole. Energy density is a relative concept, not absolute; in one reference frame it may be negative, in another positive. The exotic material can have a negative energy density as measured in the reference frame of a light beam that travels through it, but a positive energy density as measured in the wormhole's reference frame. Nevertheless, because almost all forms of matter that we humans have ever encountered have positive average energy densities in *everyone's* reference frame, physicists have long suspected that exotic material cannot exist. Presumably the laws of physics forbid exotic material, we physicists have conjectured, but just *how* the laws of physics might do so was not at all clear.